

RESEARCH INTERESTS

FinTech, Quant Finance, Real-World Model, Explainable AI, Game Theory, Social Computing, AI Ethics & Accountability, HCI.

EDUCATION

Ph.D. Student, Nanyang Technological University (NTU), Singapore	<i>2026 - Present</i>
B.Sc. in Engineering Management, Bandung Institute of Technology, Indonesia	<i>2020 - 2025</i>
Semester Exchange, International Advanced Technology Program, NTUST, Taiwan	<i>Aug 2023 - Dec 2023</i>

RESEARCH EXPERIENCE

Ph.D. Research Scientist, GIFTS Lab, Nanyang Technological University, Singapore	<i>May 2026 – Present</i>
---	---------------------------

AI Research Engineer, Singapore Management University, Singapore Hybrid	<i>Feb 2025 – Sep 2025</i>
--	----------------------------

- Built a 1,000+ question multimodal dataset benchmarking Vision-Language Models on ASEAN cultural reasoning, leading data collection, annotation, and validation under Professor Ngo Chong Wah and the SMU Multimedia Research Team.
- Co-authored “Seeing Culture: A Benchmark for Visual Reasoning and Grounding,” accepted at EMNLP 2025 Main Conference, the first benchmark stress-testing VLMs on visually grounded reasoning across nine Southeast Asian cultures.
- Developed a teacher-facing grading platform built on multimodal LLMs that auto-evaluates student submissions and surfaces learning gaps, reducing grading time and giving students faster, higher-quality feedback.
- Ran evaluation cycles on a multilingual model to improve cultural relevance, analysing failure cases and reporting weekly findings on model weaknesses that fed directly into the next training iteration.

Lab Assistant, Lab of Innovation and Corporate System Development, ITB	<i>Apr 2023 – May 2024</i>
---	----------------------------

- Built an adaptive crypto trading system on a self-hosted server, training XGBoost and other scikit-learn models on real-time market data with automated re-learning cycles that improved entry and exit timing on leveraged positions.
- Authored an undergraduate thesis applying machine learning to LQ45 trading, identifying the most predictive features driving returns across Indonesia’s top 45 stocks on Bursa Efek Indonesia.
- Investigated how consumers respond to AI-generated marketing versus human-made content using PLS-SEM and SEM-MGA, assessing the tradeoffs of AI as the creative backbone across two industry case studies.
- Taught independent classes in valuation, financial modelling, and risk assessment, and supported faculty in finance and economics courses through quiz design, grading, and student mentoring.

Research Intern, CITI Lab, NTUST (Taiwan Tech), Taipei, Taiwan	<i>Sep 2023 – Dec 2023</i>
---	----------------------------

- Took over a graduating master’s student’s Outcome Economy project under Professor Shuo-Yan Chou, building predictive maintenance and financial models for subscription-based smart home ownership.
- Modelled opportunity cost and inventory tradeoffs for the lab’s RMFS warehouse robotics project using the Newsboy framework, with the analysis folded into the team’s broader logistics pipeline.
- Co-developed demand forecasting models using the Northern Goshawk Optimization approach, running experiments and visualizations that were rolled into the lab’s downstream publications and industry partner reports.

PUBLICATIONS

Peer-Reviewed

- Burak Satar, Zhixin Ma, Patrick Amadeus Irawan, Wilfried Ariel Mulyawan, et al. “Seeing Culture: A Benchmark for Visual Reasoning and Grounding.” EMNLP 2025 Main Conference. First benchmark stress-testing Vision-Language Models on visually grounded reasoning across nine Southeast Asian cultures.

Working Papers and Technical Reports

- New Approach to Smart Home Subscription Systems in the Outcome Economy Era. NTUST CITI Lab.
- RMFS Opportunity Cost on Inventory Management. NTUST CITI Lab.

INDUSTRY EXPERIENCE

Undisclosed Prediction Market Hedge Fund (Web3 / On-chain), Remote (Full-time)	<i>Dec 2025 – Present</i>
---	---------------------------

Managing Director	<i>Feb 2026 – Present</i>
--------------------------	---------------------------

- Lead strategy and operations for a systematic portfolio across on-chain prediction markets, owning portfolio construction, position sizing, and risk frameworks across multiple concurrent strategies.
- Develop quant strategies grounded in Avellaneda-Stoikov market making, Kelly criterion sizing, Black-Scholes pricing, Markov chains, Monte Carlo simulation, and Sharpe and Sortino-based risk evaluation.

- Designed the firm's core trading infrastructure end to end: real-time on-chain data pipelines, multi-agent execution layers, dashboards, and configuration and risk controls, with a strong focus on clean execution and slippage minimisation.
- Lead a cross-institutional research team spanning NYU, NTU, and HKU across the full alpha-to-deployment loop, from hypothesis generation to live capital deployment.

Portfolio Manager *Dec 2025 – Feb 2026*

- Designed and built systematic strategies with low pairwise correlation to capture alpha across distinct sectors of the on-chain prediction market, calibrating win rate and risk-reward ratio so each strategy stayed proportional to the portfolio's overall risk budget.

Assistant Analyst, Indonesia Ministry of Public Works and Housing | Part-time

Mar – Jun 2024

- Audited Indonesian regulations affecting the ministry, mapping how legislative changes impacted policy, decision-making, and day-to-day operating procedures across the strategic planning division.
- Analysed the ARO matrix organisational structure, surfacing workflow bottlenecks and recommending realignments that clarified scope and ownership across teams.
- Drafted revised business models by reviewing existing legal and regulatory texts and identifying areas for improvement, then integrating those insights into new service delivery and stakeholder engagement plans.
- Worked directly with senior officials and external policy advisors to redefine team roles and responsibilities in line with reform objectives and legal constraints.

Digital Operations Manager, Radian Jaya, Indonesia | Hybrid

Sep 2020 – Jun 2025

- Led daily operations of the business and online store across inventory, fulfilment, customer experience, and creative content, and authored a lean returns and operations playbook that reduced friction on customer issues.
- Built a data-driven pricing and barbell strategy spanning budget and premium tiers, paired with targeted ads, SEO, and paid social campaigns that grew annual sales by roughly 20% while boosting organic traffic and overall revenue.

LEADERSHIP & ENTREPRENEURSHIP

Co-Founder and COO, Ollifant Bebe (Babycare Brand), Indonesia

Oct 2024 – Present

- Co-founded a baby and kids care brand, running consumer research with parents on hair care needs, preferences, and pain points, and drawing on my online-store background to drive organic reach and brand visibility from day one.
- Run day-to-day operations across product development, supplier coordination, branding, and go-to-market, taking new baby hair care products from concept to shelf and selling 150+ units in the first month of launch.

Founder and CEO, Miarsa (Student Housing Review Platform), Indonesia

2021 – 2022

- Founded a community-driven housing platform during COVID to fix what listing sites like Mamikos and Airbnb-style apps left out: real student experiences of the places they actually live in across multiple Indonesian cities.
- Led on-the-ground research interviewing residents on budget, day-to-day living conditions, guest policies, safety, and landlord behavior, turning raw interviews into structured listings students could trust.
- Grew the platform to 150+ active student users who relied on Miarsa to pick their housing, validating product-market fit through repeat use and word-of-mouth referrals across campuses.

TECHNICAL SKILLS

- Machine Learning: supervised and unsupervised learning, time-series forecasting, statistical modelling, regime detection, feature importance analysis, NLP, prompt engineering, model evaluation, benchmark construction.
- AI and Deep Learning: Large Language Models, Vision-Language Models, multimodal reasoning, multi-agent system design and orchestration, RAG pipelines, dataset construction, annotation pipelines.
- Data Science and Quant: algorithmic trading system development, CLOB execution layer, Web3 broker API integration, on-chain analytics, real-time market data pipelines, quantitative trading, financial modelling, sentiment analysis.
- HCI and Systems Engineering: interactive dashboard design, educational workflow AI integration, user-facing system design, self-hosted server deployment, end-to-end ML deployment, automated data pipelines.
- Programming and Tools: Python, PyTorch, scikit-learn, Pandas, NumPy, SQL, React, TypeScript, Vercel.

HONORS & AWARDS

- IISMA Scholarship: 1 of 14 awardees selected from over 12,000 national applicants for the Indonesian Government's flagship student exchange program at NTUST, Taiwan (2023).
- PMW (Program Mahasiswa Wirausaha) Entrepreneurship Award and national startup funding, awarded for founding Ticketoad, a secondhand event ticket marketplace tackling scalping and unsafe peer-to-peer resale (2022).
- 2nd Place Regional Physics (2018) and 3rd Place Regional Astronomy (2017), National Science Olympiad, DKI Jakarta.
- Silver Medalist in Chess, National Sports Week, Indonesia (2014).
- 4th Place Regional Table Tennis, National Sports Olympiad (2013).

LANGUAGES AND STANDARDIZED TESTS

English: IELTS Band 7 | Duolingo English Test: 145 Mandarin: HSK 3 Indonesian: Native GRE: Quantitative 161 | Verbal 145